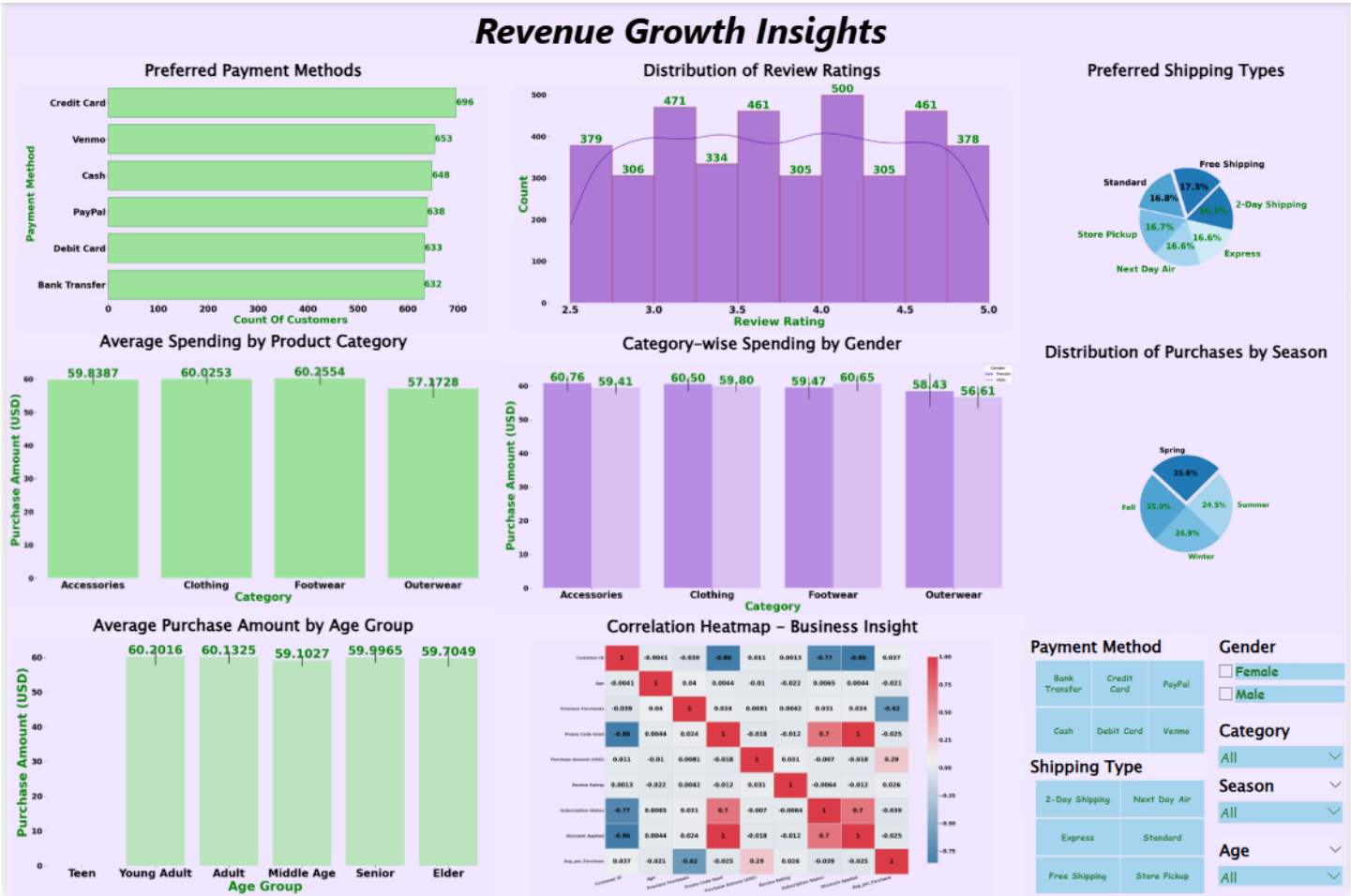


FINAL PROJECT REPORT – Customer Shopping Trends

Dashboard Title: Revenue Growth Insights

Tool Used: Google Colab + Power BI (Python Visuals)

Dataset Source: Kaggle – Customer Shopping Trends Dataset



1. Introduction

The purpose of this project is to analyze customer shopping behavior, purchasing patterns, and revenue drivers using a structured data analytics approach. The analysis focuses on identifying insights related to product categories, spending patterns, customer demographics, payment preferences, and seasonal trends. The final output is an interactive Power BI Dashboard, built using Python visuals, to support business decision-making.

2. Dataset Description

- Source: Kaggle
- File: shopping_trends.csv
- Records: 3900 rows
- Features: 19 columns
- Year / Timeline: Data collected during 2021 to 2023

Key Features:

- Customer demographics (Age, Gender, Location)
- Purchase behavior (Category, Purchase Amount, Season, Previous Purchases)
- Payment preferences
- Shipping types
- Review ratings

3. Methodology

Stage 1 – Data Loading & Exploration

- Loaded dataset in Google Colab
- Performed EDA: head(), info(), describe(), null-checks, duplicates
- Visual overview of customer behavior

Stage 2 – Data Cleaning & Preprocessing

- Converted Yes/No columns into binary format
- Created derived columns (Avg_per_Purchase)
- Removed duplicates
- Checked for skewness & outliers
- Standardized data types

Stage 3 – Visualization & Dashboard

- Built 8 key charts using Python (Matplotlib + Seaborn)
- Imported visuals into Power BI
- Designed final dashboard with slicers (Category, Gender, Season, Payment, Shipping, Age Group)

4. Key Visualizations Included

1. Purchase Amount Distribution
2. Category-wise Purchase Amount
3. Gender-wise Spending Comparison
4. Payment Method Distribution
5. Shipping Type Preference
6. Season-wise Purchase Breakdown
7. Review Rating Distribution
8. Correlation Heatmap

Each visualization was formatted using custom backgrounds, color palettes, and premium styling for readability.

5. Summary of Findings

- ❖ The shopping data shows a consistent pattern across most customer segments.
- ❖ Customers spend, on average, between \$59–\$60, regardless of age, category or gender.
- ❖ The most preferred payment method is Credit Card and customers tend to favor Free or Standard shipping options.
- ❖ Purchases are distributed almost evenly across seasons, with Spring showing a slightly higher volume.
- ❖ Customer reviews are mostly positive (4 stars being the highest count), indicating satisfaction but leaving room for improvement.

Key points in simple terms:

- Customers buy consistently across all months of the year.
- Men and women spend nearly the same amount.
- People prefer paying by Credit Card.
- Shipping cost matters — customers choose cheaper shipping.
- Reviews are good overall, but 5-star satisfaction can be improved.

6. Key Insights (3–5 Insights)

Insight 1: Stable Spending Behavior Across All Segments

Regardless of age, gender, or product category, average spending remains around \$59–\$60, showing balanced pricing and consistent customer behavior.

Insight 2: Credit Card Is the Most Preferred Payment Method

Credit Card usage leads all payment types by a large margin, suggesting strong customer trust in secure digital payments.

Insight 3: Free & Standard Shipping Are Dominant

Shipping method choice shows that customers prefer low-cost delivery options. Express or premium shipping is selected much less often.

Insight 4: Review Ratings Are Mostly Positive

Most customers provide 4-star reviews, indicating satisfaction but signaling an opportunity to improve experience further to push more ratings into 5 stars.

Insight 5: Weak Correlations Indicate Independent Behavior

The heatmap shows no strong correlations (values close to 0), meaning factors like age, gender, number of previous purchases, etc., do not strongly impact purchase amount.

7. Business Recommendations

*** Offer Free Shipping Thresholds**

Provide free shipping on orders above a certain value to increase basket size.

*** Credit Card-Based Promotions**

Introduce discounts or cashback offers with partnered banks to encourage higher-value purchases.

*** Improve Customer Review Experience**

- Offer better packaging
- Provide faster delivery
- Request follow-up feedback to boost 5-star rating volume

*** Promote Seasonal Campaigns**

Since Spring has slightly higher purchases, initiate Spring promotions early to maximize revenue.

*** Expand High-Demand Categories**

Clothing and Footwear show high demand.

Introduce more variants, bundles, or premium collections.

8. Final Story of the Dashboard :

- ❖ Customers exhibit stable spending tendencies and shop evenly across seasons.
- ❖ They prefer affordable shipping and secure payment options.
- ❖ Ratings indicate customer satisfaction, yet 5-star potential remains untapped.
- ❖ Product categories like Clothing and Footwear drive most sales, and demographic factors show limited influence.
- ❖ The business can grow further by optimizing shipping policies, improving customer experience, and launching targeted promotions aligned with payment and seasonal trends.

This dashboard allows decision-makers to visualize patterns and make informed choices for marketing, pricing, logistics, and customer retention.

9. Conclusion

- ❖ The “Revenue Growth Insights” dashboard provides a comprehensive understanding of customer behavior through clean and meaningful visualizations.
- ❖ With strategic improvements in shipping, promotion, and customer experience, businesses can significantly enhance revenue and customer satisfaction.